

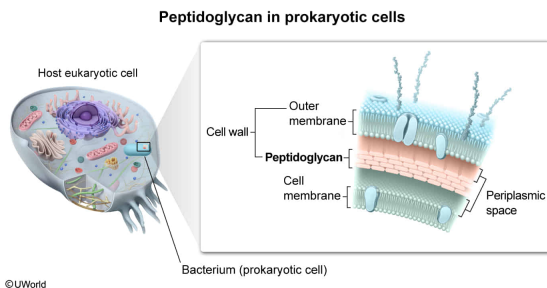
Researchers isolated peptidoglycan from a eukaryotic cell. The best explanation for this would be that:

- ✔ ☒ A. the eukaryotic cell is a host to a bacterial species containing peptidoglycan. [68%]
- ☐ B. peptidoglycan is a component of some eukaryotic cell walls. [8%]
- ☐ C. membranes within the eukaryotic cell contain peptidoglycan as a structural element. [8%]
- ☐ D. a virus with a protein coat containing peptidoglycan infected the eukaryotic cell. [14%]

Correct

67% answered correctly

Explanation:



Regardless of classification (ie, **eukaryotic or prokaryotic**), all cells are surrounded by a **plasma membrane**. In addition to this membrane, some **cell types** are surrounded by a layer known as the **cell wall**. Unlike the plasma membrane, the cell wall is rigid and provides structure to the shape of the cell it surrounds, as well as an additional layer of protection.

Nearly all prokaryotic cells have a cell wall, whereas only some eukaryotic cells do. A major component of the cell walls in bacteria is **peptidoglycan**, a polymer of sugars and short polypeptides. However, all eukaryotic cells *lack* peptidoglycan, even if they have a cell wall (ie, **plant cells**) (**Choices B and C**).

In this scenario, researchers isolated peptidoglycan from a eukaryotic cell. Because eukaryotic cells do not have peptidoglycan, this polymer must have come from another cell type located within the eukaryotic cell. For example, if the eukaryotic cell was a **host** for a **bacterial species** that contained peptidoglycan in the cell wall, the polymer could be isolated from the eukaryotic cell.

(Choice D) Viruses are obligate intracellular parasites and must **reproduce** using the molecular machinery of the host cell because they lack the ability to reproduce outside of a host cell. **Enveloped viruses** have a **phospholipid bilayer** that surrounds the viral capsid (ie, protein coat), but *no* viruses contain peptidoglycan.

Educational objective:

The primary component of bacterial cell walls is peptidoglycan. However, eukaryotic cells do not contain peptidoglycan.

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Cellular Biology